PROJECT NAME : BioPulse : Autonomous Green Ops Platform
GROUP MEMBERS : Umutcan Yavuz, Kaan Pamuk, Erdi Sinan, Ahmet Yiğit Sertel, Ata Çelik, Onur Özbek, Enes Görkem Hasgör

TASK #	PROJECT TASKS WHICH REQUIRE SOFTWARE TOOL SUPPORT
1	Digital Twin 3D Visualization
2	Backend System
3	Database Provider

SOFTWARE TOOLS FOR TASK 1: Digital Twin 3D Visualization

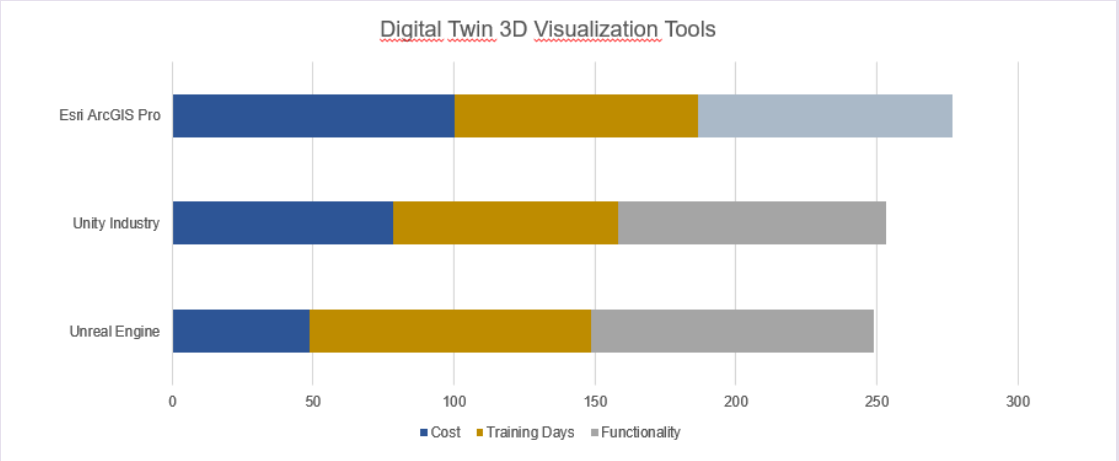
Tool Cost/Training/Functionality Data

Tool	Unreal Engine	Unity Industry	Esri ArcGIS Pro
Cost	$1850/year	$2970/year	$3800/year
Training Days	15 days	12 days	13 days
Functionality	100	95	90

Normalized Cost/Training/Functionality Data

Tool	Unreal Engine	Unity Industry	Esri ArcGIS Pro
Cost	48.7	78.2	100
Training Days	100	80	86.7
Functionality	100	95	90

Normalized Tool Graph



Which tool has been selected? Why?

Unreal Engine is selected for its industrial-grade simulation capabilities, particularly for modeling environmental physics like subterranean fluid dynamics. It provides optimal functionality at a competitive cost, significantly exceeding the performance of other options. The necessary investment in training is warranted due to the stringent accuracy needed for a safe, autonomous agricultural system.

SOFTWARE TOOLS FOR TASK 2: Single Board computers

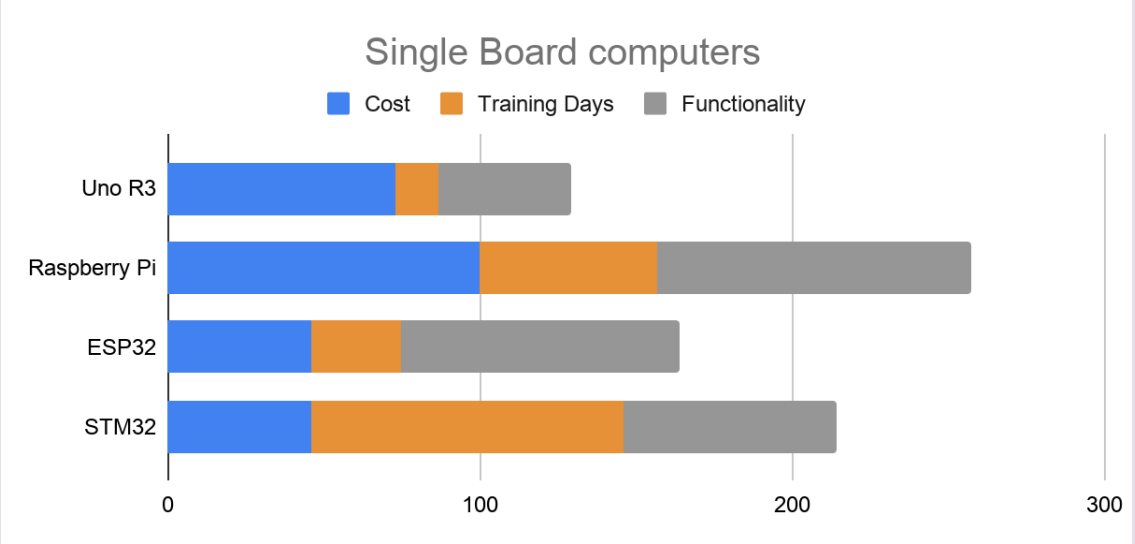
Tool Cost/Training/Functionality Data

Tool	Uno R3 Atmega328	Raspberry Pi Zero 2 W SC1176	ESP32	STM32
Cost	$10.99 / board	$15.00 / board	$6.89 / board	$6.94 / board
Training Days	2 days	8 days	4 days	14 days
Functionality	40	95	85	65

Normalized Cost/Training/Functionality Data

Tool	Uno R3 Atmega328	Raspberry Pi Zero 2 W SC1176	ESP32	STM32
Cost	73	100	46	46
Training Days	14	57	29	100
Functionality	42	100	89	68

Normalized Tool Graph



Which tool has been selected? Why?

Raspberry Pi Zero 2 W was selected as the best-fit single-board computer for the BioPulse system. Although it is slightly more expensive than the other options, it offers the highest functionality, which is important for supporting multiple sensors, real-time data handling, and communication processes.

Unlike Arduino Uno, ESP32, and STM32, Raspberry Pi has greater processing capability and can run a full operating system, which simplifies integration with databases, backend systems, and digital twin applications. Despite requiring a moderate training period, its strong performance and flexibility make it the most effective and balanced choice for our project.

SOFTWARE TOOLS FOR TASK 3: Database Provider

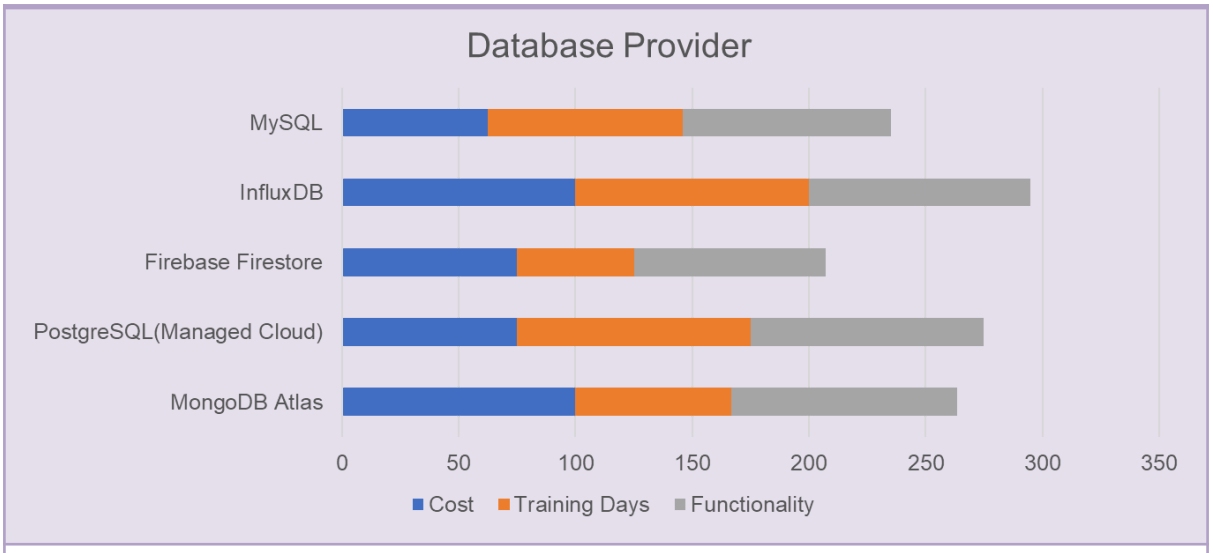
Tool Cost/Training/Functionality Data

Tool	MySQL	InfluxDB	Firebase Firestore	PostgreSQL	MongoDB Atlas
Cost	$300/year	$480/year	$360/year	$360/year	$480/year
Training Days	5 days	6 days	3 days	6 days	4 days
Functionality	85	90	78	95	92

Normalized Cost/Training/Functionality Data

Tool	MySQL	InfluxDB	Firebase Firestore	PostgreSQL	MongoDB Atlas
Cost	62.5	100	75	75	100
Training Days	83.3	100	50	100	66.7
Functionality	89.5	94.7	82.1	100	96.8

Normalized Tool Graph



Which tool has been selected? Why?

MongoDB Atlas was selected for the BioPulse system because it is well-suited for handling flexible and real-time sensor data. The system processes semi-structured data such as soil moisture, temperature, humidity, minerals such as calcium, magnesium, phosphorus and alerts, making a NoSQL approach more suitable than relational databases. Compared to other tools, MongoDB provides a good balance between functionality, flexibility, and ease of use, making it the most practical choice for our project.